Fiscal Year:	FY 2008	**Task Last Updated:**	FY 11/02/2007
PI Name:	Goree, John Ph.D.		
Project Title:	Self-Structuring in Dusty Plasmas (NNX07AD22G)		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	52242-1403	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03-C
Start Date:	03/01/2007	**End Date:**	02/28/2010
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Chait, Arnon	**Contact Phone:**	216-433-3558
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Piel, Alexander (University of Kiel) Melzer, Andre (University of Greifswald,)		
Grant/Contract No.:	NNX07AD22G		
Performance Goal No.:			
Performance Goal Text:			
	This fundamental research is an experimental study of dusty plasma physics. Ground-based laboratory research are carried out in preparation for future microgravity flight experiments using European instruments on the International Space Station ISS. A dusty plasma consists of particles of solid matter suspended in an ionized gas. Micron-size particles acquire a large negative charge of typically 10,000 elementary charges. Due to inter-particle repulsion, they arrange themselves in structure with solid or liquid-like properties, similar to a colloidal crystal. With particles of this size, gravity is a dominant force, causing particles to sediment to a thin layer at the bottom of the plasma. Under microgravity conditions, particles fill a 3D volume and the suspension self-organizes. We will study: • Structural		

Task Description:	features. These include the boundaries of the particle suspension, including a void at the plasma's center and vortices at its edges. • Dynamical behavior exhibited by the dusty plasma. These include waves, instabilities, and transport. This dynamical behavior will be studied at an atomistic level, using video micrography to observe the motion of microspheres which serve as an analog for atoms in normal molecular matter. Experimental methods will include direct imaging of particle suspensions using video micrography and particle tracking, and manipulation of particles using the radiation-pressure force of powerful cw lasers. Tasks for the next three years include planning for the ISS experiment and carrying out ground-based laboratory experiments that will serve the dual purpose of developing scientific methods to be used in future flight experiments as well as providing an immediate scientific return. The future spaceflight microgravity experiments are expected to be performed using European instruments on ISS. NASA will not be expected to provide any flight hardware. In these future microgravity experiments, video cameras will be used to image not only the particle suspension as in previous experiments, but also spectral lines from the plasma glow. The spectral measurements will yield data on the plasma conditions that produce the ion drag. Additionally, using laser manipulation, a known force will be applied to particles to measure the force balance in situ. These space-flight experiments on ISS will be carried out in cooperation with German scientists.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	For the reporting period 3/1/07 - 10/31/07 Ground-based laboratory experiments were carried out, manuscripts were submitted to journals, and collaboration with German co-I's was carried out, as detailed below. Two experiments were performed: (1) A study of the melting/freezing transition of a two-dimensional (2D) suspension of dust particles in a plasma. Using a laser to heat the suspension we melt a crystalline structure. Then, turning the laser off abruptly, we allow it to freeze. This freezing process goes through two phases, where the first appears to be a supercooled liquid, and the second is a slow relaxation of the supercooled liquid to a crystalline state. This is a preliminary experiment, which will be reported in an upcoming American Physical Society meeting. The experiment will be repeated before publishing the results. (2) A study of flows in a three-dimensional (3D) suspension of dust particles in a plasma. The effects of gravity are mostly offset, in this ground-based experiment, using the thermophoretic force, which is a force applied to dust particles immersed in a gas with a temperature gradient. Starting with a suspension in a liquid state, we used a powerful laser to apply forces to particles, stirring up the suspension, and we observed both the resulting flow and a change in the shape of the suspension. This is a preliminary experiment, which will be reported in an upcoming American Physical Society meeting. The experiment will be repeated before publishing the results. During this eight-month period (3/1 - 10/31/07) we published three papers in refereed journals, based mostly on work performed during the precedecessor grant Self-Structuring in Dusty Plasmas NNC05GA43G. We submitted an additional three papers, which are currently under review by journals. We gave one invited and two contributed talks at conferences. Collaborative research with the German PIs consisted of a meeting with a researcher from University of Kiel. We worked on the draft of a manuscript we plan to submit to a journal. This paper will report a ground-based experiment that was carried out in 2006 in Iowa by the Kiel researcher.
Bibliography Type:	Description: (Last Updated: 01/23/2025)
Articles in Peer-reviewed Journals	Liu B, Goree J. "Superdiffusion in two-dimensional Yukawa liquids." Phys Rev E Stat Nonlin Soft Matter Phys. 2007 Jan;75(1 Pt 2):016405. PMID: 17358266 , Jan-2007
Articles in Peer-reviewed Journals	Feng Y, Goree J, Liu B. "Accurate particle position measurement from images." Rev Sci Instrum. 2007 May;78(5):053704. PMID: 17552822 , May-2007
Articles in Peer-reviewed Journals	Donkó Z, Hartmann P, Goree J. "Shear viscosity of strongly-coupled two-dimensional Yukawa liquids: experiment and modeling." Modern Physics Letters B, 2007 Sep;21(10):1357-76. http://dx.doi.org/10.1142/S0217984907013948 , Sep-2007